

# Introduction to Programming II

## project log

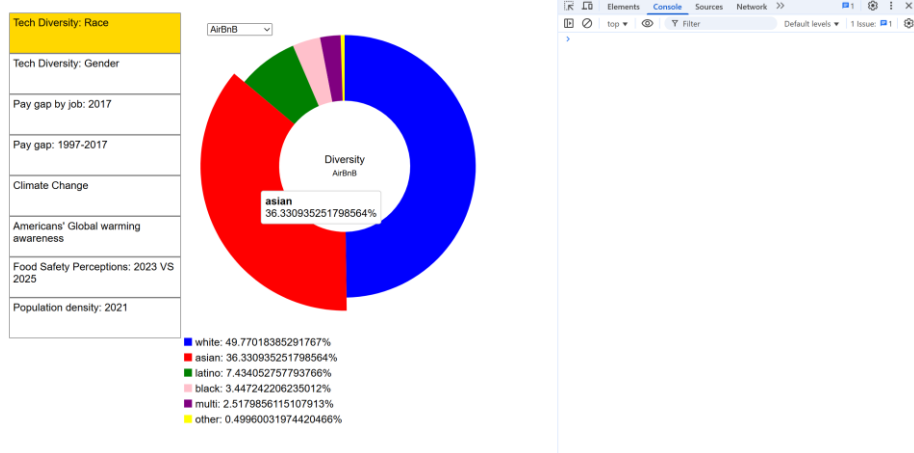
**Project title:** Coursework – Data Visualisation

**Topic:** Week 19 and 20: Usability tests

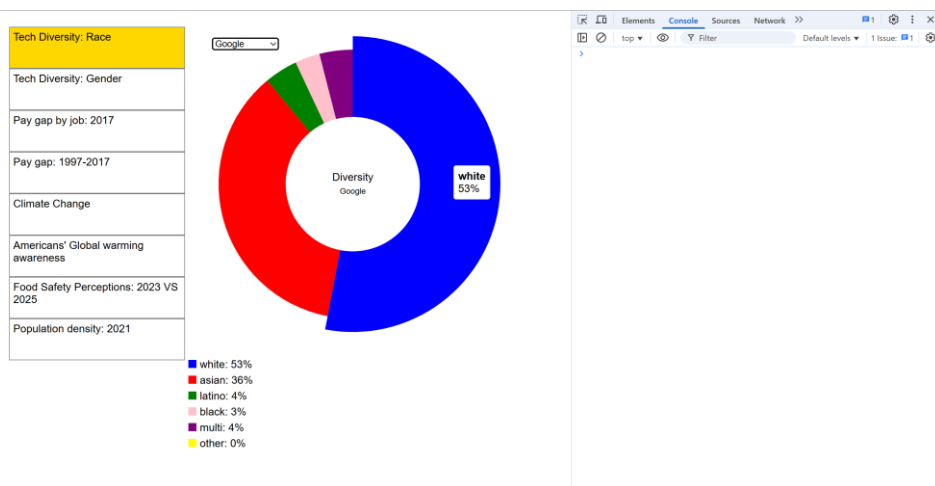
**What progress have you made on this topic?**

Stability test: I went through all data visualisations to check for consistent performance.

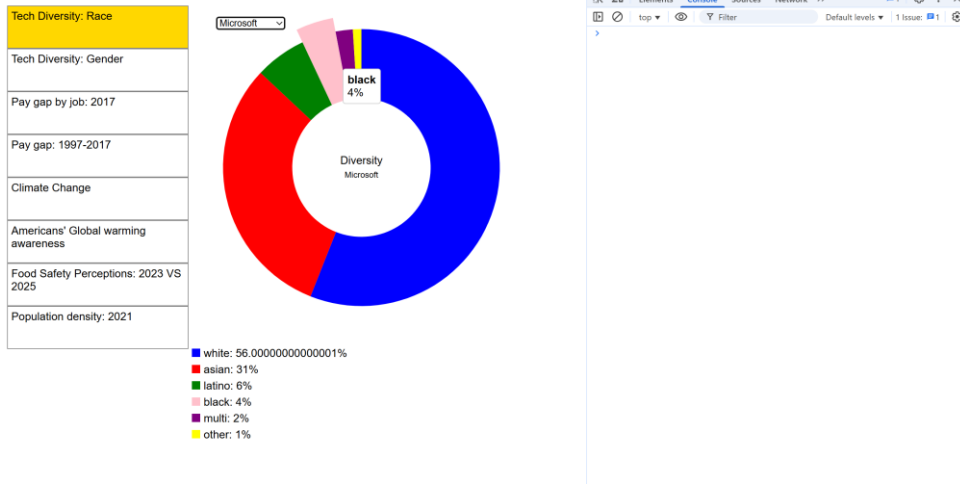
### 1. Tech Diversity: Race



Pic 1: Company: AirBnB



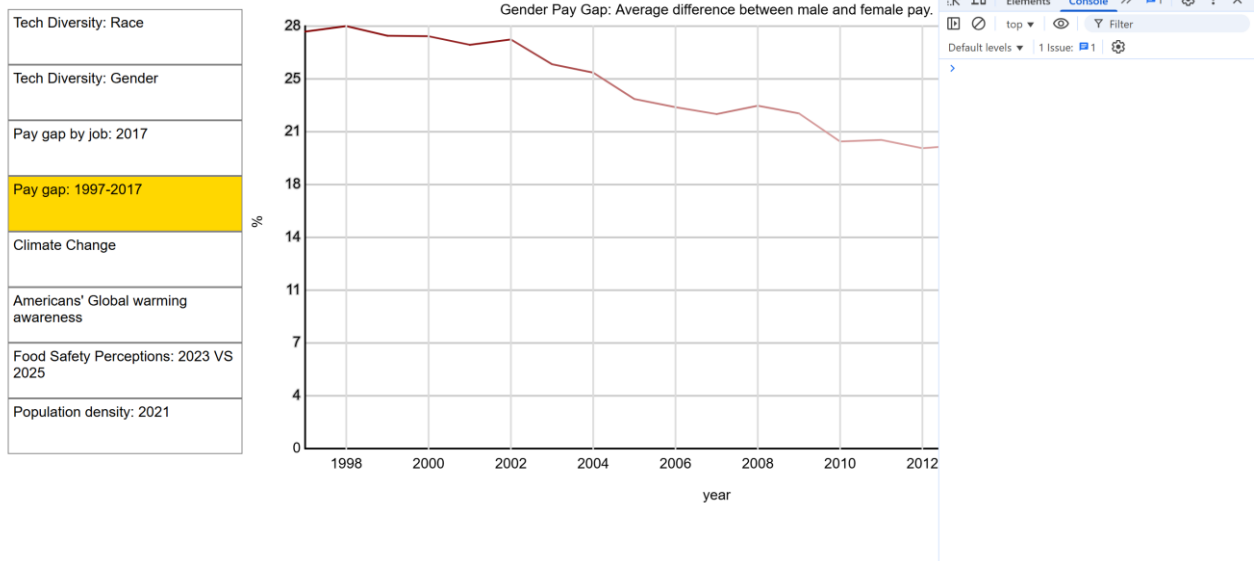
Pic 2: Company: Google



Pic 3: Company: Microsoft

| CHECKLIST |                                                           |
|-----------|-----------------------------------------------------------|
| ✓         | Chart renders correctly on load                           |
| ✓         | Legends are positioned and readable                       |
| ✓         | No console errors or warnings                             |
| ✓         | Tooltips, hover states, or interactivity work as intended |
| ✓         | Performance remains smooth                                |

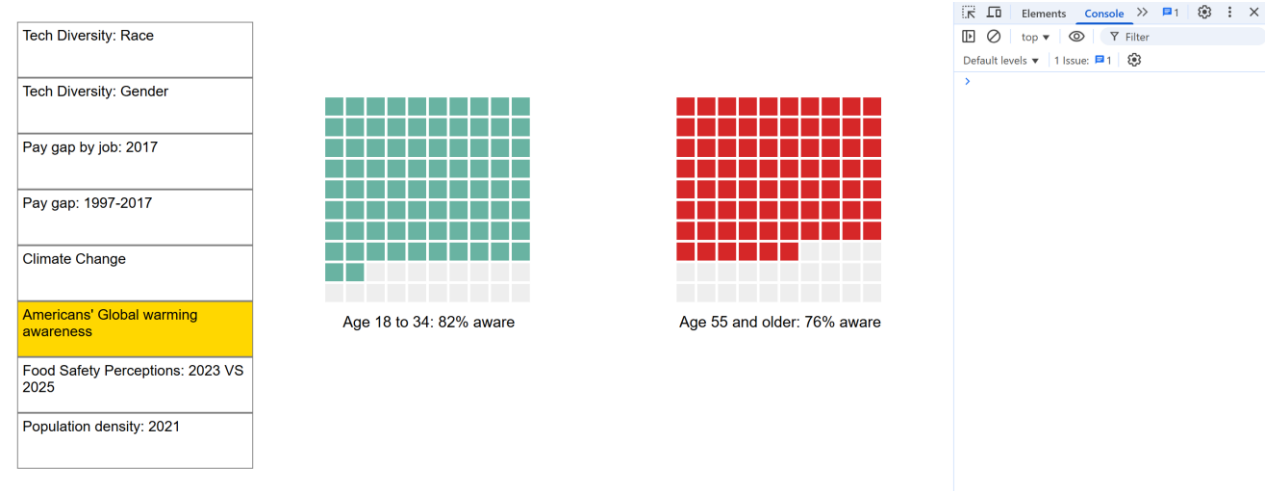
## 2. Pay gap: 1997-2017



Pic 4: Colour intensity line graph

| CHECKLIST |                                   |
|-----------|-----------------------------------|
| ✓         | Graph renders correctly on load   |
| ✓         | No console errors or warnings     |
| ✓         | Colour intensity work as intended |
| ✓         | Performance remains smooth        |

3. American’s Global warming awareness

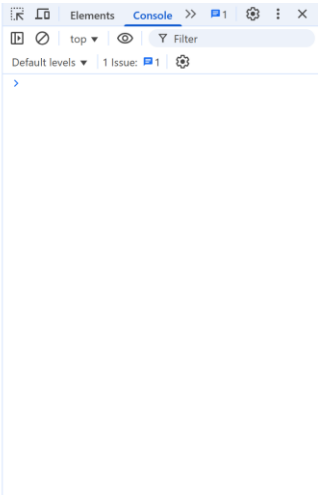
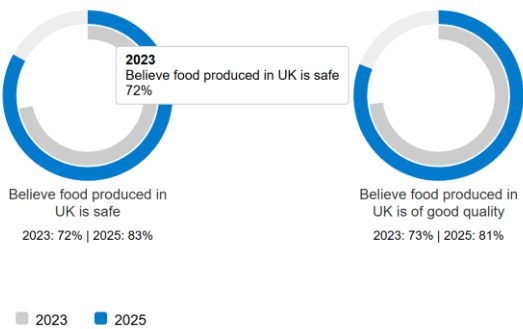


Pic 5: Waffle Charts

| CHECKLIST |                                     |
|-----------|-------------------------------------|
| ✓         | Graph renders correctly on load     |
| ✓         | No console errors or warnings       |
| ✓         | Legends are positioned and readable |
| ✓         | Performance remains smooth          |

4. Food Safety Perception: 2023 VS 2025

|                                       |
|---------------------------------------|
| Tech Diversity: Race                  |
| Tech Diversity: Gender                |
| Pay gap by job: 2017                  |
| Pay gap: 1997-2017                    |
| Climate Change                        |
| Americans' Global warming awareness   |
| Food Safety Perceptions: 2023 VS 2025 |
| Population density: 2021              |

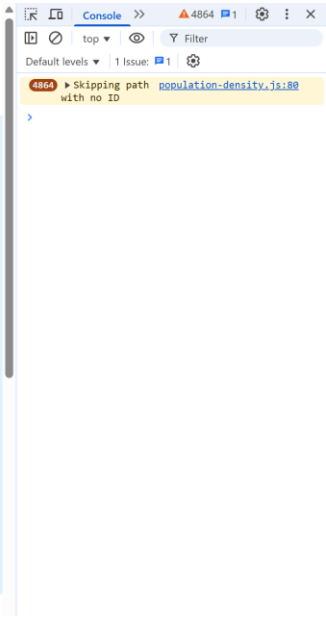
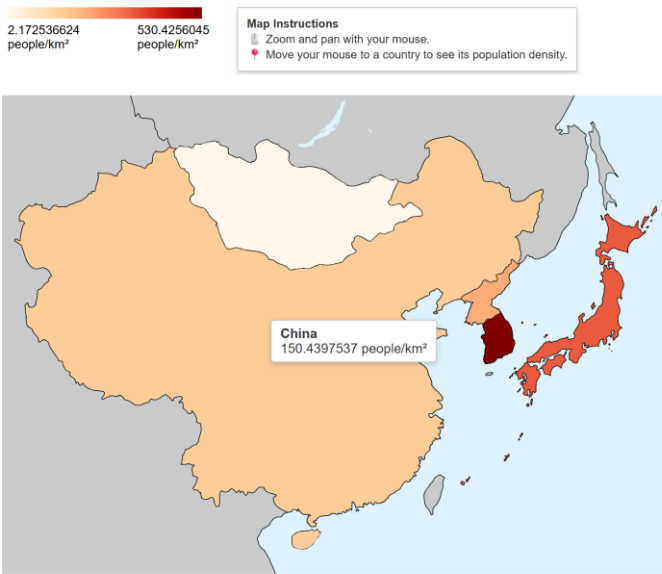


Pic 6: Circle Diagrams

| CHECKLIST |                                                           |
|-----------|-----------------------------------------------------------|
| ✓         | Chart renders correctly on load                           |
| ✓         | Legends are positioned and readable                       |
| ✓         | No console errors or warnings                             |
| ✓         | Tooltips, hover states, or interactivity work as intended |
| ✓         | Performance remains smooth                                |

5. Population density: 2021

|                                       |
|---------------------------------------|
| Tech Diversity: Race                  |
| Tech Diversity: Gender                |
| Pay gap by job: 2017                  |
| Pay gap: 1997-2017                    |
| Climate Change                        |
| Americans' Global warming awareness   |
| Food Safety Perceptions: 2023 VS 2025 |
| Population density: 2021              |



Pic 7: Heat Map

| CHECKLIST |                                                           |
|-----------|-----------------------------------------------------------|
| ✓         | Chart renders correctly on load                           |
| ✓         | Legends and instructions are positioned and readable      |
| ✓         | No console errors or warnings                             |
| ✓         | Tooltips, hover states, or interactivity work as intended |
| ✓         | Performance remains smooth                                |

All extended data visualisations functioned as intended, with no errors detected.

I also conducted usability surveys with three participants using the System Usability Scale (SUS). Their responses provided valuable insights for the system's overall usability and user experience.

| Question                                                                                      | 1                                | 2                     | 3                     | 4                     | 5                                |
|-----------------------------------------------------------------------------------------------|----------------------------------|-----------------------|-----------------------|-----------------------|----------------------------------|
|                                                                                               | Strongly disagree                |                       |                       |                       | Strongly agree                   |
| 1. I think that I would like to use this system frequently.                                   | <input type="radio"/>            | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input checked="" type="radio"/> |
| 2. I found the system unnecessarily complex.                                                  | <input checked="" type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>            |
| 3. I thought the system was easy to use.                                                      | <input type="radio"/>            | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input checked="" type="radio"/> |
| 4. I think that I would need the support of a technical person to be able to use this system. | <input checked="" type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>            |
| 5. I found the various functions in this system were well integrated.                         | <input type="radio"/>            | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input checked="" type="radio"/> |
| 6. I thought there was too much inconsistency in this system.                                 | <input checked="" type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>            |
| 7. I would imagine that most people would learn to use this system very quickly.              | <input type="radio"/>            | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input checked="" type="radio"/> |
| 8. I found the system very cumbersome to use.                                                 | <input checked="" type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>            |
| 9. I felt very confident using the system.                                                    | <input type="radio"/>            | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input checked="" type="radio"/> |
| 10. I needed to learn a lot of things before I could get going with this system.              | <input checked="" type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>            |

Pic 8: Response 1

| Question                                                                                      | 1                                | 2                     | 3                     | 4                                | 5                     |
|-----------------------------------------------------------------------------------------------|----------------------------------|-----------------------|-----------------------|----------------------------------|-----------------------|
|                                                                                               | Strongly disagree                |                       |                       |                                  | Strongly agree        |
| 1. I think that I would like to use this system frequently.                                   | <input type="radio"/>            | <input type="radio"/> | <input type="radio"/> | <input checked="" type="radio"/> | <input type="radio"/> |
| 2. I found the system unnecessarily complex.                                                  | <input checked="" type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>            | <input type="radio"/> |
| 3. I thought the system was easy to use.                                                      | <input type="radio"/>            | <input type="radio"/> | <input type="radio"/> | <input checked="" type="radio"/> | <input type="radio"/> |
| 4. I think that I would need the support of a technical person to be able to use this system. | <input checked="" type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>            | <input type="radio"/> |
| 5. I found the various functions in this system were well integrated.                         | <input type="radio"/>            | <input type="radio"/> | <input type="radio"/> | <input checked="" type="radio"/> | <input type="radio"/> |
| 6. I thought there was too much inconsistency in this system.                                 | <input checked="" type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>            | <input type="radio"/> |
| 7. I would imagine that most people would learn to use this system very quickly.              | <input type="radio"/>            | <input type="radio"/> | <input type="radio"/> | <input checked="" type="radio"/> | <input type="radio"/> |
| 8. I found the system very cumbersome to use.                                                 | <input checked="" type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>            | <input type="radio"/> |
| 9. I felt very confident using the system.                                                    | <input type="radio"/>            | <input type="radio"/> | <input type="radio"/> | <input checked="" type="radio"/> | <input type="radio"/> |
| 10. I needed to learn a lot of things before I could get going with this system.              | <input checked="" type="radio"/> | <input type="radio"/> | <input type="radio"/> | <input type="radio"/>            | <input type="radio"/> |

☐ Dark Mode

Instructions

Enter your scores, per question, to calculate total SUS score. A score should be given for all questions.

If you do not feel that you can respond to a particular question then mark it at the centre point of the scale.

Result

SUS Score: 100

By conventional standards a score of 100 is considered to be above average.

This score is considered a Grade A.

This score may be described as *best imaginable*.

☐ Dark Mode

Instructions

Enter your scores, per question, to calculate total SUS score. A score should be given for all questions.

If you do not feel that you can respond to a particular question then mark it at the centre point of the scale.

Result

SUS Score: 87.5

By conventional standards a score of 87.5 is considered to be above average.

This score is considered a Grade B.

This score may be described as *excellent*.

Pic 9: Response 2

| Question                                                                                      | 1                                | 2                                | 3                     | 4                                | 5                                |
|-----------------------------------------------------------------------------------------------|----------------------------------|----------------------------------|-----------------------|----------------------------------|----------------------------------|
|                                                                                               | Strongly disagree                |                                  |                       |                                  | Strongly agree                   |
| 1. I think that I would like to use this system frequently.                                   | <input type="radio"/>            | <input type="radio"/>            | <input type="radio"/> | <input checked="" type="radio"/> | <input type="radio"/>            |
| 2. I found the system unnecessarily complex.                                                  | <input checked="" type="radio"/> | <input type="radio"/>            | <input type="radio"/> | <input type="radio"/>            | <input type="radio"/>            |
| 3. I thought the system was easy to use.                                                      | <input type="radio"/>            | <input type="radio"/>            | <input type="radio"/> | <input type="radio"/>            | <input checked="" type="radio"/> |
| 4. I think that I would need the support of a technical person to be able to use this system. | <input type="radio"/>            | <input checked="" type="radio"/> | <input type="radio"/> | <input type="radio"/>            | <input type="radio"/>            |
| 5. I found the various functions in this system were well integrated.                         | <input type="radio"/>            | <input type="radio"/>            | <input type="radio"/> | <input type="radio"/>            | <input checked="" type="radio"/> |
| 6. I thought there was too much inconsistency in this system.                                 | <input checked="" type="radio"/> | <input type="radio"/>            | <input type="radio"/> | <input type="radio"/>            | <input type="radio"/>            |
| 7. I would imagine that most people would learn to use this system very quickly.              | <input type="radio"/>            | <input type="radio"/>            | <input type="radio"/> | <input checked="" type="radio"/> | <input type="radio"/>            |
| 8. I found the system very cumbersome to use.                                                 | <input checked="" type="radio"/> | <input type="radio"/>            | <input type="radio"/> | <input type="radio"/>            | <input type="radio"/>            |
| 9. I felt very confident using the system.                                                    | <input type="radio"/>            | <input type="radio"/>            | <input type="radio"/> | <input checked="" type="radio"/> | <input type="radio"/>            |
| 10. I needed to learn a lot of things before I could get going with this system.              | <input checked="" type="radio"/> | <input type="radio"/>            | <input type="radio"/> | <input type="radio"/>            | <input type="radio"/>            |

☐ Dark Mode

**Instructions**  
Enter your scores, per question, to calculate total SUS score. A score should be given for all questions.  
If you do not feel that you can respond to a particular question then mark it at the centre point of the scale.

**Result**  
**SUS Score: 90**  
By conventional standards a score of 90 is considered to be above average.  
This score is considered a Grade B.  
This score may be described as *excellent*.

Pic 10: Response 3

All 3 responses had SUS scores above average, indicating a generally positive perception of the system's usability. This suggests that the system is perceived as usable and well-designed by its target users.

**What problems have you faced and were you able to solve them?**

No critical issues were encountered. All extensions and feature enhancements were successfully integrated and validated through stability and usability testing.

**What are you planning to do over the next few weeks?**

In this final week, I'm consolidating my work, refining documentation, and ensuring all deliverables meet quality standards.

**Are you on target to successfully complete your project? If you aren't on target, how will you address the issue?**

I am on target to successfully complete my project. Following a comprehensive round of stability, usability, and performance testing, all features have been validated and the final version is ready for submission.

